

VR Motor Control Diagnostic Tool

Operator and Commissioning Guide — Qt 6 desktop diagnostic for CiA 402 motor drives

v1.0 Powered by `vr-mc-sdk` · built against Qt 6 · transports: UDP-multicast CAN (`hal_can_udp`) + ZLG USB-CANFD (`hal_can_zlg`)

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1. Overview & architecture

The diagnostic is a standalone Qt 6 desktop application that reads and writes CiA 402 motor drives over a `hal_can` transport (UDP-multicast for loopback / bench work, ZLG USB-CANFD for real drives). It wraps the `master_mgr` + `motor_drive_interface` C layers from `vr-mc-sdk` in a dedicated Qt worker thread, surfaces live TPDO telemetry in a QtCharts scope, and exposes every commissioning knob through modeless dialogs that can sit next to the main window while the operator works.

Architecture — vr-mc-diagnostic runtime

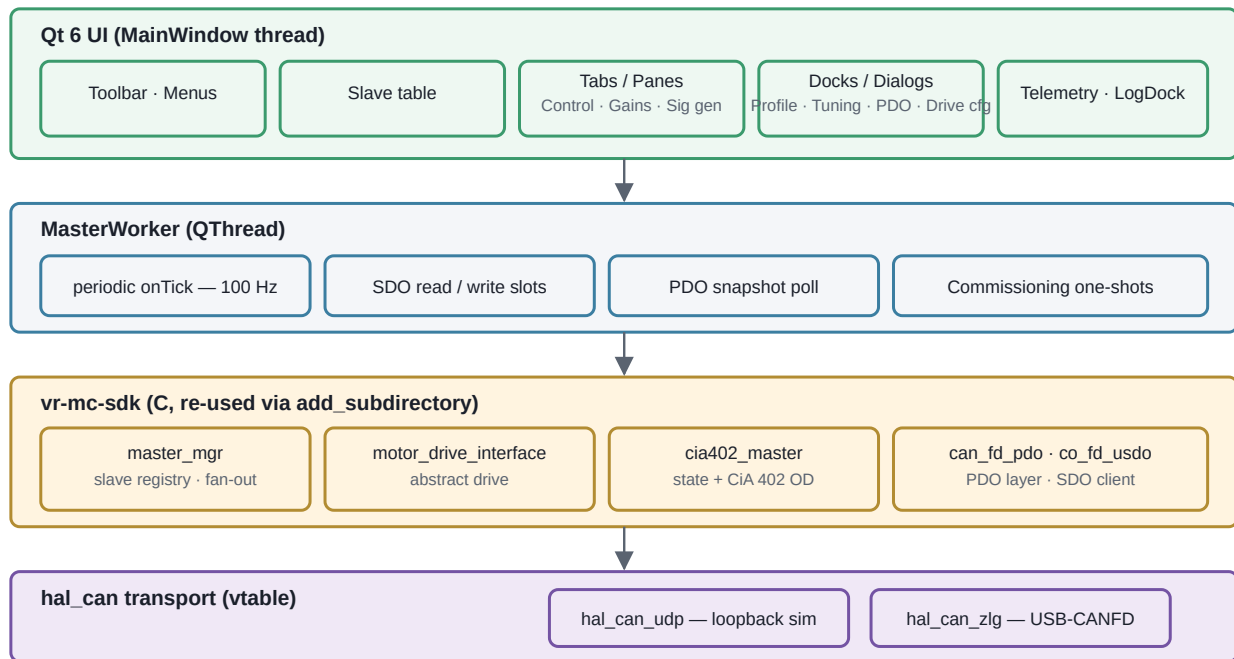


Fig. 1 — Runtime architecture. The Qt UI thread never calls the C SDK directly; **MasterWorker** owns the master stack on its own **QThread** and communicates via queued signals/slots.

The layering is intentional:

- **UI thread** builds widgets, routes user input, paints charts. Never blocks on SDO.
- **Worker thread** owns `master_mgr_t`, runs the periodic refresh, and performs all SDO read/write work. Emits a snapshot signal at 100 Hz.
- **SDK layer** (C code) — `master_mgr` fans out to per-slave `motor_drive_interface`; the CiA 402 implementation talks to `co_fd_usdo` (SDO) and `can_fd_pdo` (PDO) over a `hal_can` vtable.
- **Transport layer** — `hal_can_udp` or `hal_can_zlg` concrete, chosen at connect time.

Thread safety. Every exchange between the UI and the worker goes through Qt signals with `Qt::QueuedConnection`. No raw pointer handoffs, no shared-state races. Even the periodic refresh grabs the worker's mutex, so an Apply press and a refresh tick serialise naturally.

2. Install & first run

2.1 Prerequisites

- Qt 6.2 or newer (Core, Gui, Widgets, Charts).
- CMake 3.16+, a C++17 compiler.
- Linux host — `hal_can_udp` uses BSD sockets + multicast; `hal_can_zlg` loads `libcontrolcanfd.so`.
- For the ZLG path: a USB-CANFD adapter plugged in.

2.2 Build from source

```
git clone .../vr-hand
cd vr-hand/vr-mc-diagnostic
cmake -S . -B build
cmake --build build --target vr_mc_diagnostic -j
./build/vr_mc_diagnostic
```

2.3 Loopback bring-up (no hardware)

The SDK ships a CiA 402 simulator you can use to bring the diagnostic up without a real drive. Run it in one terminal, then launch the diagnostic in another:

```
./vr-mc-sdk/build/app/motor_drive_cia402 --udp --node 5
./vr-mc-diagnostic/build/vr_mc_diagnostic --auto-connect
```

The simulator emits TPDO1 at 2 kHz on multicast `239.192.0.42:23400`; `--auto-connect` wires the diagnostic to the same group and pulls up node id 5.

3. Connecting to drives

3.1 Opening the Connect dialog

From the menu: **Connection** → **Connect...** From the keyboard: `Ctrl+K`. The dialog offers two transports:

Transport	Fields	When to use
UDP multicast	Group (<code>239.192.0.42</code>), Port (<code>23400</code>)	Loopback against the <code>motor_drive_cia402</code> simulator, multi-host bench.
ZLG USB-CANFD	Channel index, arbitration bitrate, data-phase bitrate	Real drives on a USB-CANFD adapter.

Both transports take **First node id** and **Count**. The diagnostic sequentially registers that range of slaves into `master_mgr`, each gaining a row in the table.

3.2 After Connect

- Status bar flips to *connected* with the slave count.
- Table fills: *state* = *offline* transitions to the live statusword-decoded string once TPDO1 starts flowing.
- **Reconnect** (`Ctrl+R`) replays the last config without re-opening the dialog.
- **Disconnect** tears everything down idempotently — worker thread stays alive for the next connect.

4. Main window tour

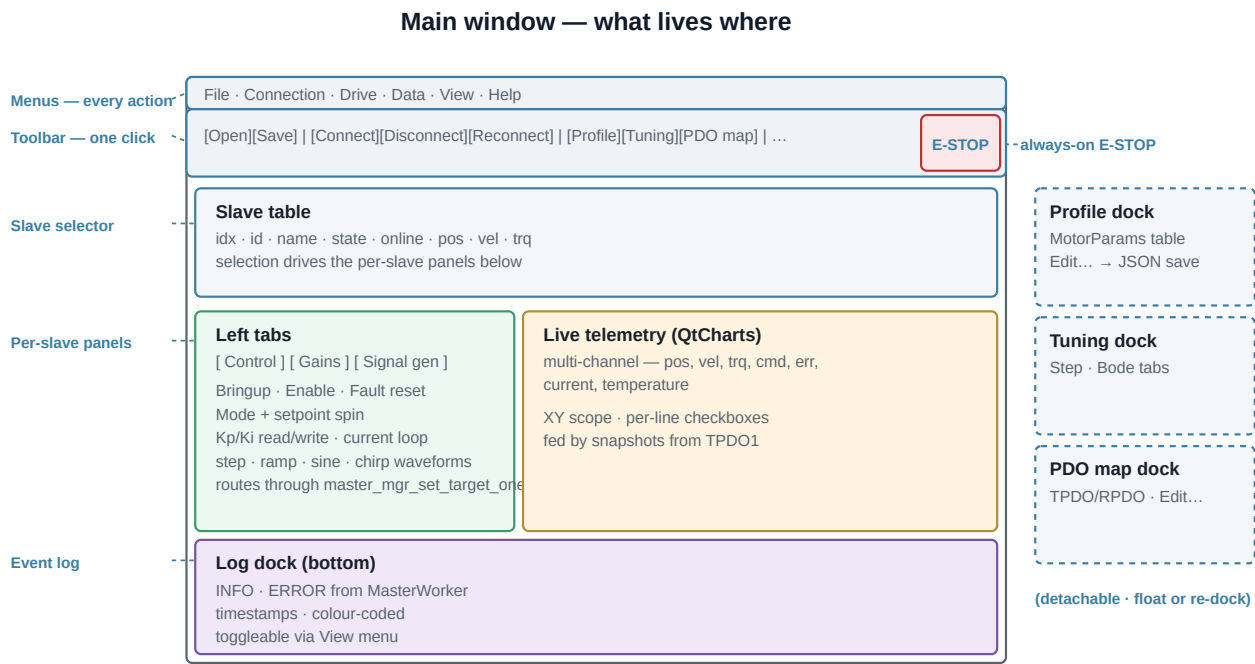


Fig. 2 — Annotated layout. The frame is fixed; the left tabs and telemetry chart expand to fill; detachable panes dock on the right or float as top-level windows.

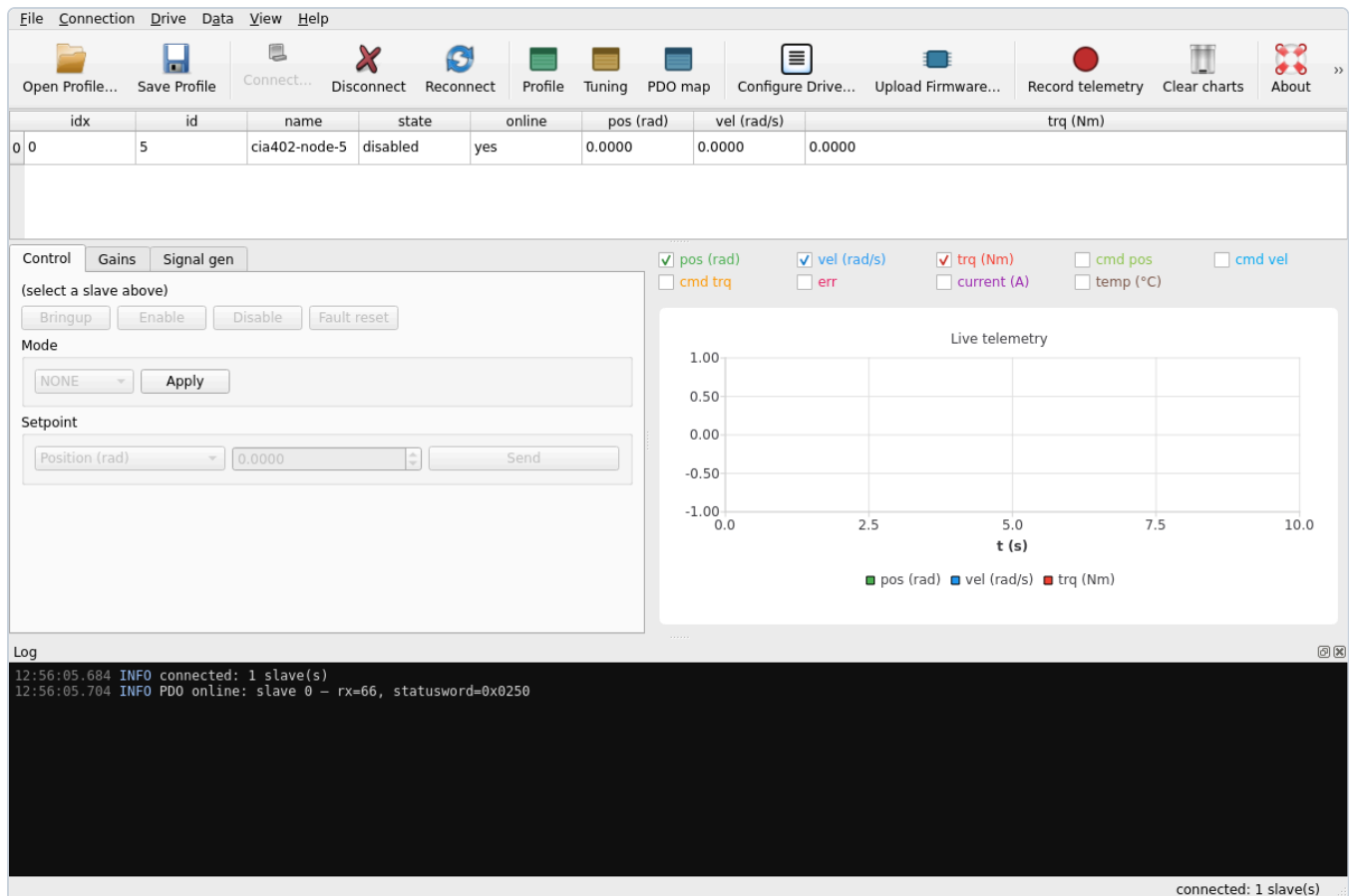


Fig. 3 — Main window connected to the loopback simulator. Slave `cia402-node-5` is registered and online; the log confirms PDO traffic (`rx=66`, `statusword=0x0250`).

4.1 Toolbar groups

Left to right, the toolbar is grouped by intent:

1. **File** — Open / Save motor profile.
2. **Connection** — Connect... / Disconnect / Reconnect.
3. **View** — toggle the Profile, Tuning, and PDO map docks. Same actions also live in the **View** menu.
4. **Drive** — Configure Drive... (Ctrl+Shift+C) + Upload Firmware.
5. **Data** — Record telemetry + Clear charts.
6. **Help** — About.
7. **E-STOP** — always on the right, always clickable, shortcut `Space` .

4.2 Slave table

One row per slave. Columns: `idx` · `id` · `name` · `state` · `online` · `pos` · `vel` · `trq`. Selecting a row routes it to every per-slave panel — Control, Gains, Signal gen, Telemetry, and the dialogs that need a target.

5. Left-tab panels

5.1 Control tab

Drives the CiA 402 state machine. Each button issues a controlword transition through `master_mgr_*_one` :

CiA 402 drive state machine — what Control-tab buttons do

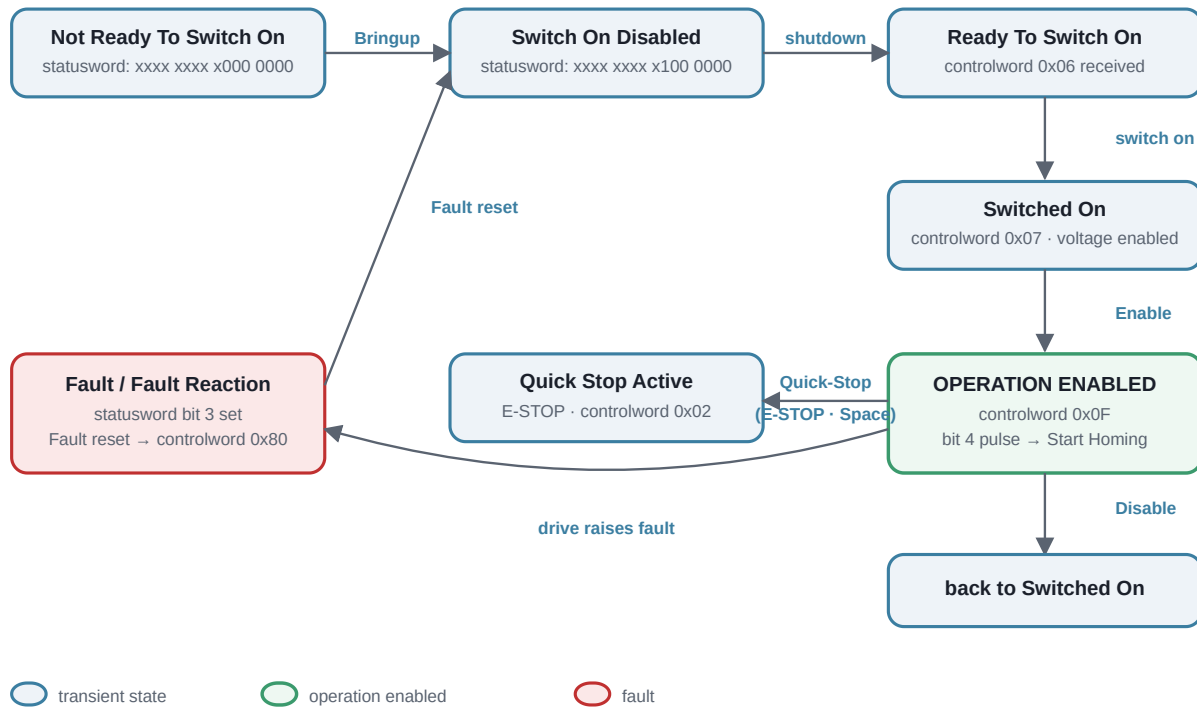


Fig. 4 — CiA 402 state machine. Control-tab buttons pulse the controlword; telemetry reports the resulting statusword back through TPDO1.

- **Bringup** — walks the drive from *Not Ready to Switch On* to *Switched On*.
- **Enable** — final transition into *Operation Enabled*. Only state that accepts motion commands.
- **Disable** — back to *Switched On*, torque off.
- **Fault reset** — pulses controlword bit 7 to clear a fault; drive lands in *Switch On Disabled*.

Mode selector picks *none · torque · velocity · position*; **Setpoint** sends the numeric target. Both route through `master_mgr_set_target_one`.

5.2 Gains tab

idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000

Control | **Gains** | Signal gen

Loop | Kp | Ki

Current | 0.000000 | 0.000000 | Read | Apply

Velocity | 0.000000 | 0.000000 | Read | Apply

Position | 0.000000 | 0.000000 | Read | Apply

Read all

Live telemetry

pos (rad) | vel (rad/s) | trq (Nm)

Log

```
12:56:11.537 INFO connected: 1 slave(s)
12:56:11.555 INFO PDO online: slave 0 - rx=64, statusword=0x0250
```

connected: 1 slave(s)

Fig. 5 — Per-loop Kp/Ki editor. **Read** pulls the drive's current values via SDO, **Write** pushes edited values back.

Three loops (*current · velocity · position*) with independent Kp and Ki entry. Reads round-trip through `motor_drive_intf_get_gain`; writes go through `motor_drive_intf_set_gain`.

5.3 Signal generator

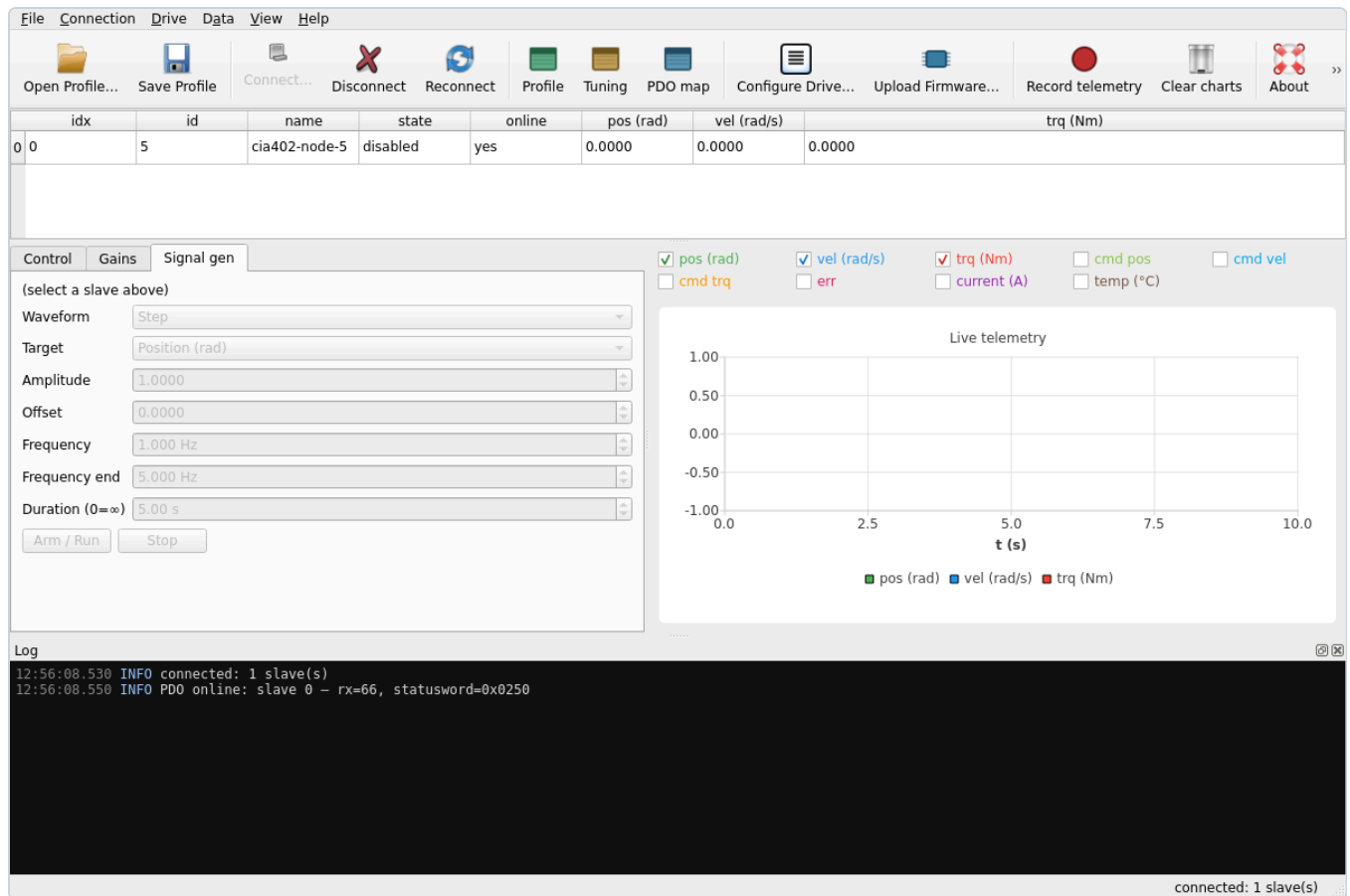


Fig. 6 — Signal generator. Waveforms: constant, step, ramp, sine, chirp. Target channel, amplitude, offset, frequency, duration, and update rate are all operator-tunable.

- **Constant / Step** — one-shot setpoint.
- **Ramp** — slope = amplitude / duration.
- **Sine** — pure tone at the given frequency.
- **Chirp** — log-swept sine from frequency start → end across duration; feeds the Bode view.

6. Detachable panes

Profile, Tuning, and PDO map are created as `QDockWidget`s — hidden and floating by default so the main window stays uncluttered. Clicking their toolbar button (or the **View** menu entry) surfaces the dock; the operator can float or re-dock it to any edge of the main window.

6.1 Motor profile

The screenshot shows the motor control software interface. The 'Motor profile' dock is re-docked to the right, displaying a table of motor parameters. The table has columns for 'Paramet', 'Value', and 'Unit'. The parameters listed are:

Paramet	Value	Unit
Type	PM...	
Pole pairs	4	
R_s	0.5	Ω
L_{s_d}	0.0015	H
L_{s_q}	0.0015	H
Rat...	0.05	Wb
Ine...	5e-05	kg-m ²
Rat...	1000	rad/s
Rat...	24	V
Min volt...	12	V
Max volt...	36	V
Rat...	2	A
Max cur...	6	A
Stall	0	A

The 'Edit...' button is visible at the bottom of the table. The main control panel on the left includes tabs for 'Control', 'Gains', and 'Signal gen'. The 'Control' tab is active, showing a 'Setpoint' section with a dropdown menu set to 'Position (rad)' and a value of '0.0000'. The 'Live telemetry' graph shows position (rad) in green, velocity (rad/s) in blue, and torque (Nm) in red. The log window at the bottom shows the following messages:

```
12:56:17.539 INFO connected: 1 slave(s)
12:56:17.559 INFO PDO online: slave 0 - rx=66, statusword=0x0250
```

Fig. 7 — Profile dock re-docked to the right. Read-only 15-row table covers nameplate params (type, pole pairs, R_s , L_{sd} , L_{sq} , rated flux, inertia) + voltage and current limits.

Edit... opens a modal form editor. **File** → **Save Profile** persists to JSON.

6.2 Tuning

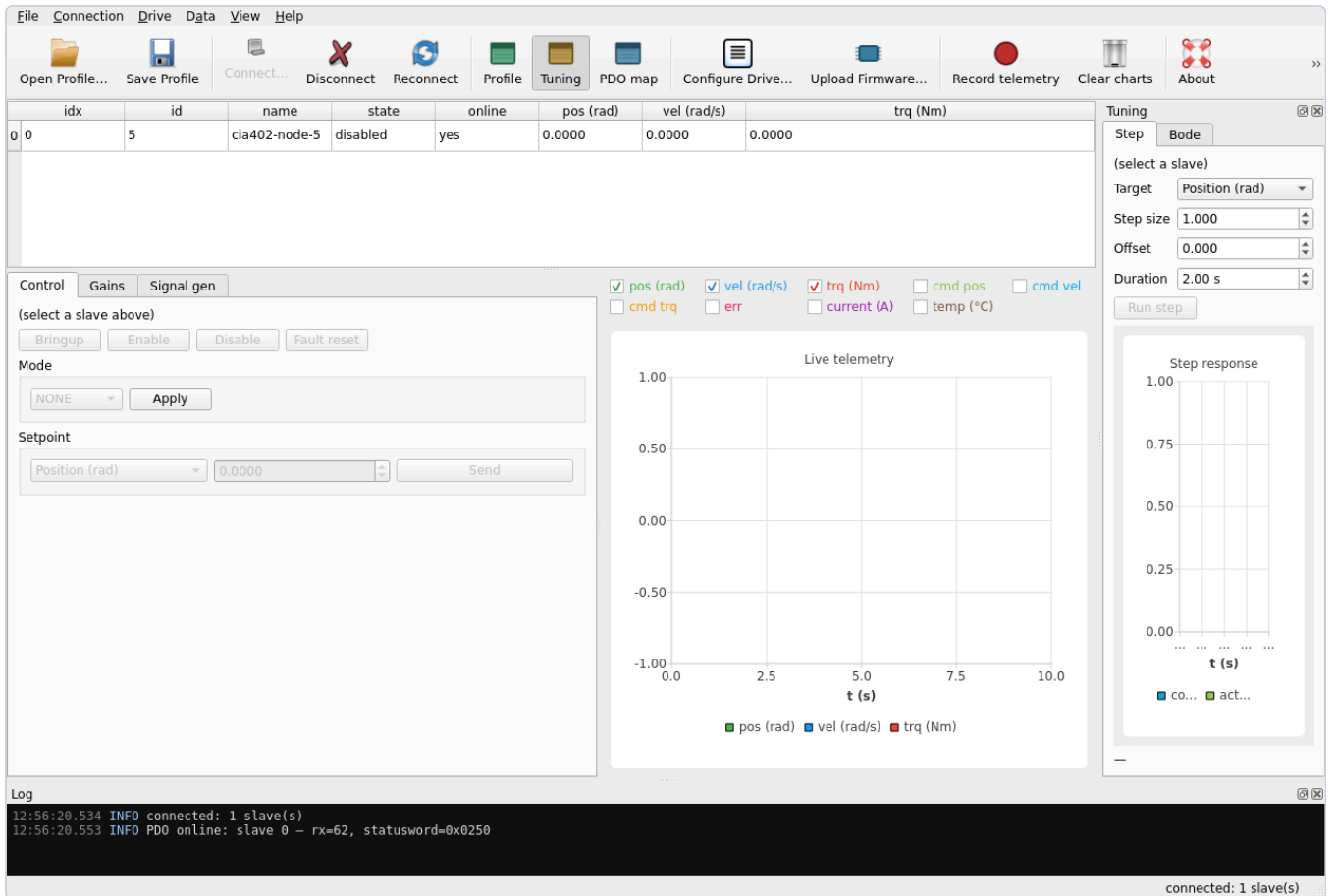


Fig. 8 — Tuning dock with the Step sub-tab active. Set target channel, step size, offset, and duration; **Run step** triggers the generator and the pane overlays command vs. actual.

- **Step** — reports overshoot (%), 10–90 % rise time, and steady-state error.
- **Bode** — log-sweep chirp + Goertzel-correlation yields magnitude (dB) and phase (°) vs. frequency on a log axis.

6.3 PDO map

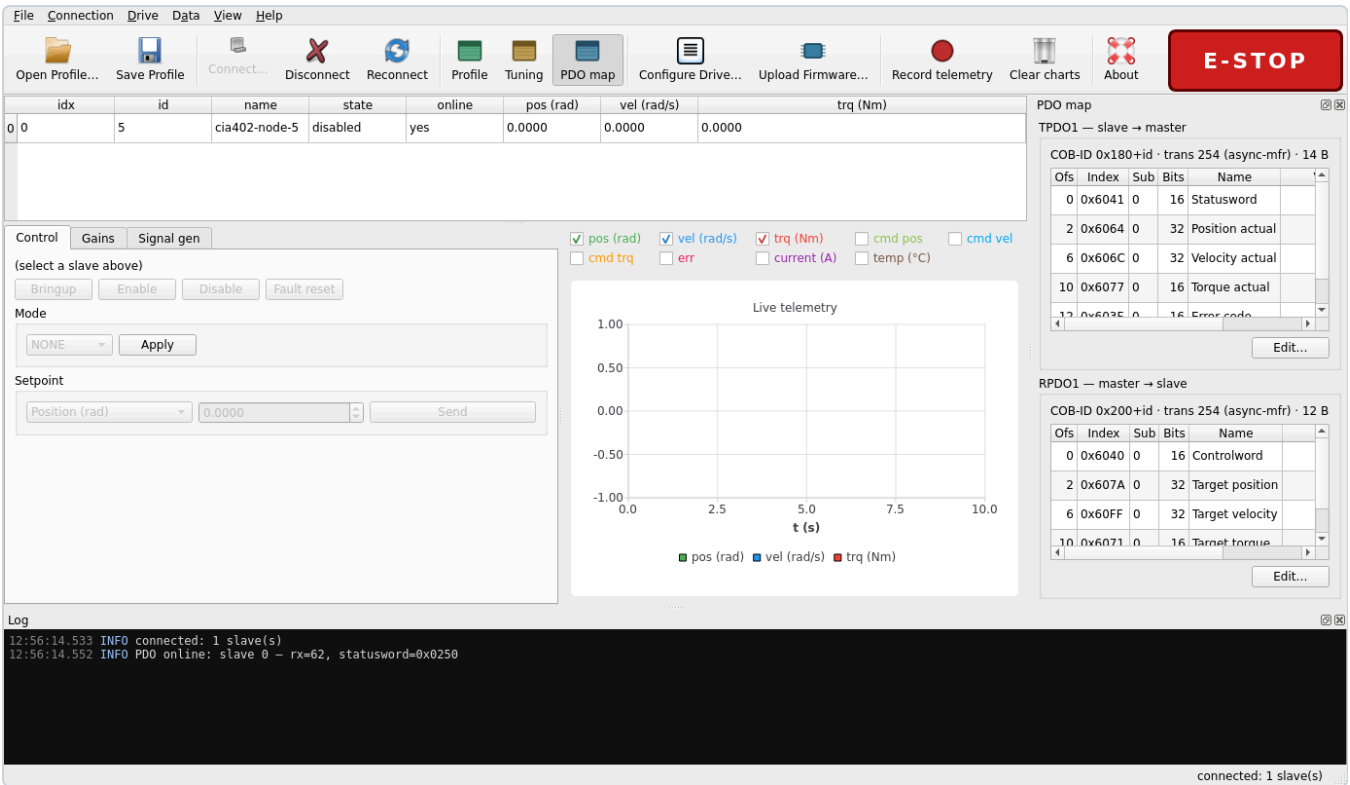


Fig. 9 — PDO map dock. Two sections (TPDO1 slave → master, RPDO1 master → slave). Each row shows offset, OD index, sub, bits, name, and a live-decoded value tracked off the latest snapshot.

7. Configure Drive

Drive → **Configure Drive...** (**Ctrl+Shift+C**) opens a modeless tabbed dialog targeted at the currently-selected slave. The dialog kicks off an SDO read on open so the form reflects what's on the drive, not a blank default.

Slave #0 — cia402-node-5

Homing
Motion profile
Protection
Encoder

Method (0x6098)0  no homing

Home offset (0x607C)0 counts

Switch search speed (0x6099:1)100 cnt/s

Zero search speed (0x6099:2)10 cnt/s

Homing acceleration (0x609A)1000 cnt/s²

Start HomingZero encoder here

Statusword 0x0250 — in progress...

Read from drive
Apply

OK
Cancel

Fig. 10 — Configure Drive dialog, Homing tab. Every field labeled with its CiA 402 OD coordinate. Status banner at the bottom decodes statusword bits 10 / 12 / 13 in real time while homing runs.

7.1 Homing tab

Field	OD	Notes
Method	0x6098	Combobox with standard CiA 402 methods (0, 1, 2, 7, 11, 17, 18, 23, 27, 33–35, 37, -1, -2).
Home offset	0x607C	Origin shift applied by the drive after homing completes.
Switch search speed	0x6099:1	Fast approach until a switch / limit edge is detected.
Zero search speed	0x6099:2	Slow creep onto the actual zero (index / hard stop).
Homing acceleration	0x609A	Ramp to the search speeds.

Two commissioning one-shots live below the form:

- **Start Homing** — writes mode = 6 to 0x6060, then pulses controlword bit 4 on 0x6040 (0x000F → 0x001F). Status banner decodes statusword bits 10, 12, 13 live.

- **Zero encoder here** — reads position_actual (0x6064), writes it into home_offset (0x607C), and updates the form field so Apply will persist it.

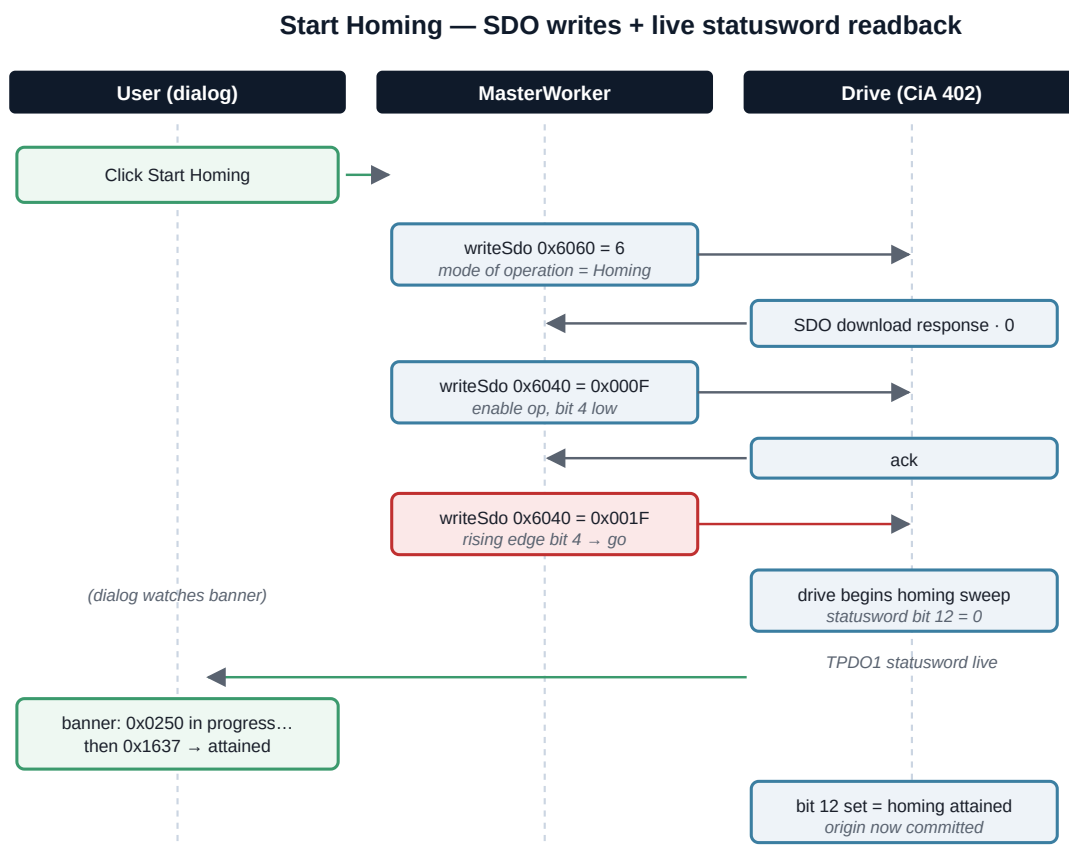


Fig. 11 — Start Homing message sequence. Three SDO writes kick the drive; progress comes back through the existing TPDO1 statusword stream, so no polling is needed.

7.2 Motion profile tab

- Profile velocity — 0x6081
- Profile acceleration — 0x6083
- Profile deceleration — 0x6084
- Quickstop deceleration — 0x6085

These set the trapezoidal profile the drive uses when it moves under *Profile Position / Profile Velocity* modes (CiA 402 modes 1 / 3). Quick-stop decel is what the drive uses when E-STOP issues controlword 0x02.

7.3 Protection tab

- Max following error — 0x6065 (counts); faults the drive if position error exceeds it.
- Following error window — 0x6066 (ms); how long the error has to persist.
- Position limit min / max — 0x607D:1,2 ; hard software travel limits.
- Max motor speed — 0x6080 (counts/s).
- Max torque — 0x6072 (% of rated).

Zero torque here reads torque_actual (0x6077) and writes it into torque_offset (0x60B2). Lets the current loading read as zero — useful when taring against gravity.

7.4 Encoder tab

- Resolution — `0x608F:1` (counts per revolution).
- Gear numerator / denominator — `0x6091:1,2`.
- Feed constant numerator / denominator — `0x6092:1,2`.

7.5 Buttons

- **Read from drive** — batch SDO upload over the full field set. Missing OD entries are reported in the log but don't abort the batch.
- **Apply** — batch SDO download. Partial failures land in the log with per-field reason codes + a warning dialog.
- **OK** — Apply + close. **Cancel** — discard.

8. Commissioning workflow

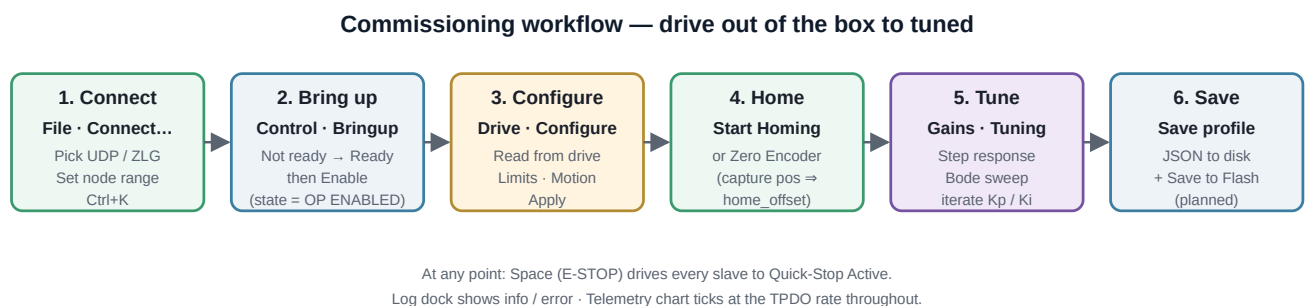


Fig. 12 — End-to-end commissioning — from cold connect to tuned and saved, in six steps.

1. **Connect** (`Ctrl+K`). Pick transport, set node range, Connect. Confirm the row in the table turns *online* = yes.
2. **Bring up**. Select the row, open **Control**, click **Bringup**. State walks to *Switched On*. Click **Enable**. State: *Operation Enabled*.
3. **Configure**. **Drive** → **Configure Drive...** → **Read from drive**. Review the Protection tab: pos limits and max torque should reflect the mechanical envelope. Apply.
4. **Home**. On the Homing tab, either click **Start Homing** for a method-driven sweep, or drive the joint to a known position and click **Zero encoder here** to capture it as the origin.
5. **Tune**. Open the **Gains** tab, **Read** the starting Kp/Ki. Use the **Tuning** dock: run a step response, read overshoot / rise time, adjust. For rotational dynamics, run a Bode sweep via the chirp waveform.
6. **Save**. **File** → **Save Profile** writes the edited nameplate JSON. Save to Flash via **Drive** → **Save Config to Flash** (planned) persists runtime params on the drive side.

Always have an exit. Keep one hand near `Space` (E-STOP) the first time you enable a new drive. The diagnostic has no opinion about your mechanics — if limits aren't set right, motion can escape the safe envelope within a control cycle.

9. Tuning workflows

9.1 Step response

1. Drive must be in *Operation Enabled* and in the target mode (e.g. *Profile Position*).
2. Tuning dock → **Step** sub-tab. Pick target channel (position / velocity / torque), step size, offset, duration.
3. Click **Run step**. The diagnostic overlays cmd vs. actual on the chart and computes overshoot (%), rise time (10 → 90 %), and steady-state error.
4. Iterate: increase Kp until overshoot becomes visible, back off 20 %, then add Ki to kill the steady-state error.

9.2 Bode sweep

1. Tuning dock → **Bode** sub-tab. Set frequency start / end (typically 1 Hz → 200 Hz) and sweep duration (30 s is generous).
2. **Run sweep**. The diagnostic fires a chirp waveform; for each frequency the signal generator emits, a Goertzel correlator extracts in-phase and quadrature components of the measured output.
3. Read magnitude crossover (−3 dB) and phase margin from the plot. Loop-shape accordingly.

10. PDO mapping — read, edit, apply

The diagnostic's PDO layer uses two `hal_can` endpoints — one for SDO, one for PDO. Keeping them separate avoids the single-consumer `hal_can_set_cb` from shadowing either traffic:

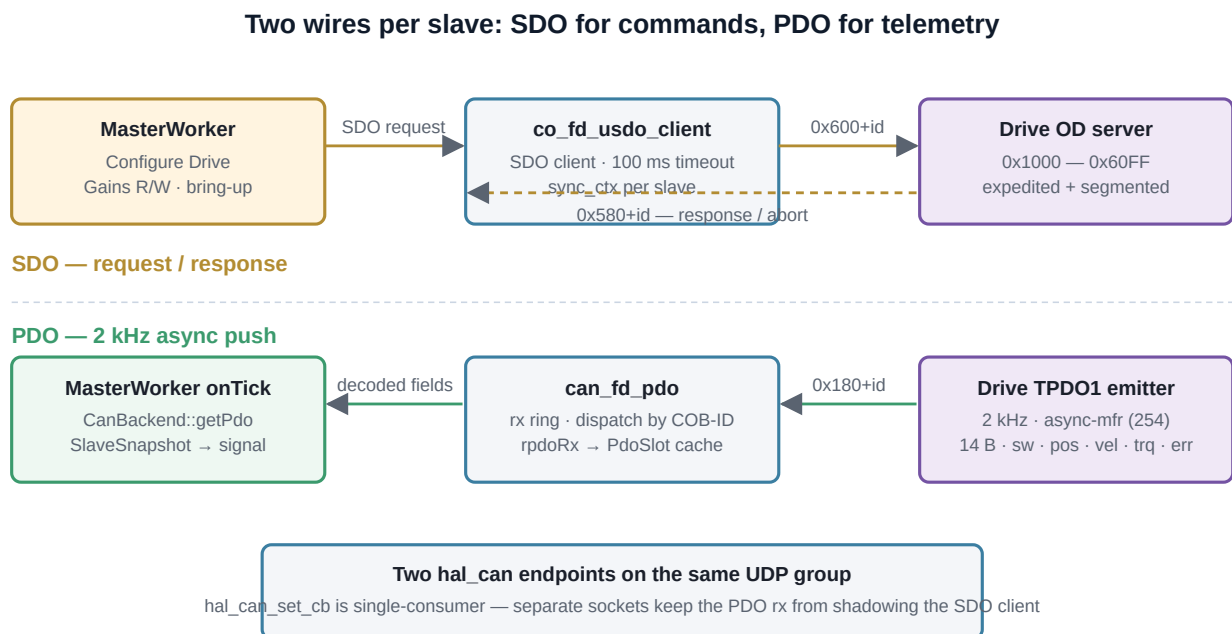


Fig. 13 — SDO and PDO take different paths through the stack. SDO is request/response with 100 ms timeout; PDO is a 2 kHz push into a ring buffer drained by the worker.

Editing the mapping happens via the PDO map dock:

1. Click **Edit...** under TPDO1 or RPDO1.

2. Add entries from the CiA 402 object catalog (Controlword, Statusword, Target / Actual Position, Velocity, Torque, Error Code, DC link voltage, ...). Rearrange with Up / Down. Each entry contributes `bit_length / 8` bytes to the frame.
3. Click **OK**. The diagnostic runs the CiA 301 §7.2.2.2 remap dance via SDO:

CiA 301 §7.2.2.2 — PDO remap dance (TPDO1 example)

Every step is an SDO write from MasterWorker to the slave.

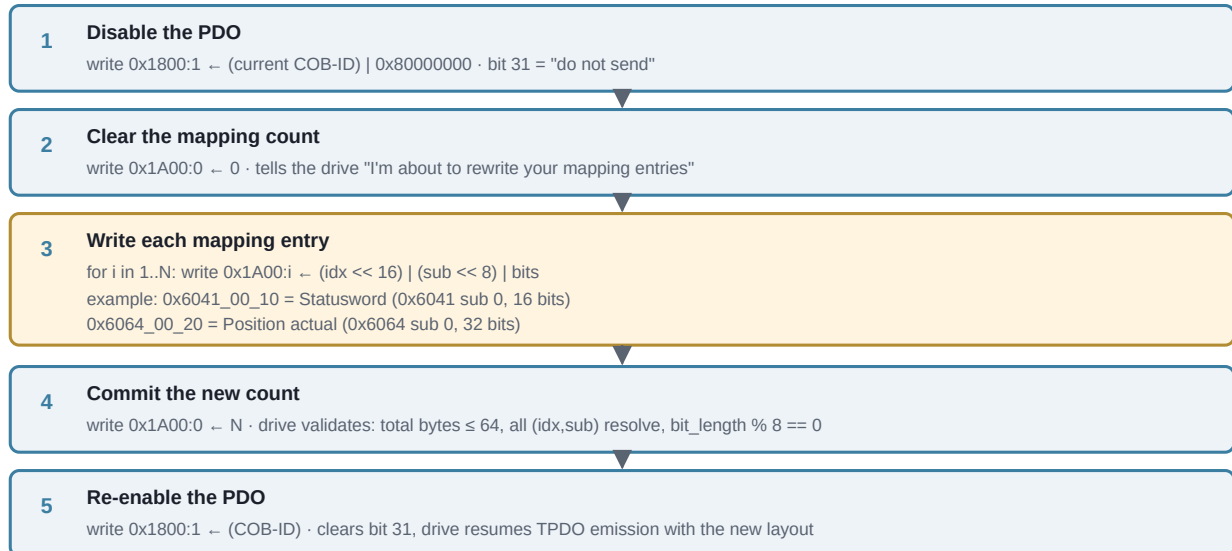


Fig. 14 — CiA 301 remap dance. Every arrow is one SDO write. The drive validates the resulting mapping on the step-4 commit and rejects anything that overflows 64 B or references a missing OD entry.

Note. Against the in-tree simulator the PDO config objects (`0x1600` / `0x1A00` and friends) are not exposed, so the remap aborts per-step with SDO abort code `0x06020000` . The log surfaces the per-step reason. On a real CiA 402 drive the sequence completes and TPDO layout changes take effect on the next PDO emission.

11. Telemetry recording & export

- **Record telemetry** (`Ctrl+Shift+R`) — toggled from the Data menu or toolbar. Currently a placeholder that writes to the log; CSV-per-snapshot is the planned evolution.
- **Export telemetry CSV...** — dumps the currently buffered chart data (what the Telemetry pane has accumulated) to a timestamped CSV.
- **Clear charts** — empties the scope buffer.

12. Emergency stop

The large red **E-STOP** button is pinned to the right of the toolbar and is always clickable. Shortcut `Space` is application-wide — it fires even when a setpoint spin box or name field has focus. On click:

1. Any running signal generator is stopped.
2. `master_mgr_disable_all` walks every slave into *Switch-On Disabled* (via controlword `0x00`).

3. The log records "E-STOP: disabled N slave(s)".

Re-enabling a slave after E-STOP is just the normal Bringup → Enable sequence on the Control tab.

13. Keyboard shortcuts

Combo	Action
Ctrl+K	Connect...
Ctrl+R	Reconnect (replay last config)
Ctrl+E	Edit motor profile
Ctrl+Shift+C	Configure Drive...
Ctrl+Shift+R	Toggle telemetry recording
Ctrl+O	Open profile...
Ctrl+S	Save profile
Space	E-STOP (application-wide)
F1	Documentation

14. Troubleshooting

Symptom	Likely cause	Check
Configure Drive: "all fields aborted"	Drive doesn't expose the CiA 402 config objects (simulator, minimal drives).	Use a real drive, or leave the dialog for display only.
CO_FD_USDO_CLIENT: Session timeout	Drive not responding. Wrong node id, bus down, transport mis-selected, or PDO callback eating SDO frames.	Verify node id, transport; check the two UDP sockets came up.
PDO map Value column stays —	No TPDO received — drive not emitting, COB-ID mismatch, or TPDO not enabled.	Log should show "PDO online" once per slave; if not, sniff the bus.
E-STOP disabled	Not connected yet — no slaves to disable.	Connect first; it lights up automatically.
Start Homing does nothing visible	Drive not in Operation Enabled before bit 4 pulse.	Hit Enable on Control first.
Floating dock disappeared	Window moved off-screen.	Toggle it from the toolbar / View menu to re-show.

Appendix A — Transports in detail

A.1 UDP multicast (`hal_can_udp`)

The diagnostic opens two sockets per connection: one for SDO traffic, one for PDO. Both bind `0.0.0.0:23400` on the loopback interface with `SO_REUSEADDR` so they coexist in the same process. Each socket joins the `239.192.0.42` multicast group; `IP_MULTICAST_LOOP` stays on so other local sockets see our sends. Every frame rides a 12-byte header:

```
[ sender_tag : 4 ][ cob_id : 4 ][ dlc : 1 ][ flags : 1 ][ rsvd : 2 ]
```

The sender tag filters self-echoes — each socket generates a random tag at create time.

A.2 ZLG USB-CANFD (`hal_can_zlg`)

`libcontrolcanfd.so` is opened via `dlopen()` at runtime; the shared object ships in `src/backends/zlg/lib/` and is found through the RPATH. One endpoint per channel; PDO is not yet wired on this transport (V1 is SDO-only).

Appendix B — CiA 402 OD quick reference

Objects the diagnostic reads or writes:

Index	Sub	Field	Where
0x1010	1	Store parameters	Drive · Save to Flash (planned)
0x1011	1	Restore defaults	Drive · Load from Flash (planned)
0x1400+n	1	RPDO comm — COB-ID	PDO remap · disable bit
0x1600+n	0..N	RPDO mapping	PDO mapping editor
0x1800+n	1	TPDO comm — COB-ID	PDO remap · disable bit
0x1A00+n	0..N	TPDO mapping	PDO mapping editor
0x603F	0	Error code	TPDO1 · PDO map
0x6040	0	Controlword	Control tab · Homing
0x6041	0	Statusword	TPDO1 · decoded in banners
0x6060	0	Mode of operation	Control tab · Homing
0x6064	0	Position actual	TPDO1 · Zero encoder here
0x6065	0	Max following error	Configure Drive · Protection
0x6066	0	Following error timeout	Configure Drive · Protection
0x606C	0	Velocity actual	TPDO1
0x6072	0	Max torque	Configure Drive · Protection
0x6077	0	Torque actual	TPDO1 · Zero torque here
0x607A	0	Target position	RPDO1
0x607C	0	Home offset	Homing tab · Zero encoder
0x607D	1,2	Position limits	Protection
0x6080	0	Max motor speed	Protection
0x6081	0	Profile velocity	Motion profile
0x6083	0	Profile acceleration	Motion profile
0x6084	0	Profile deceleration	Motion profile
0x6085	0	Quickstop deceleration	Motion profile
0x608F	1	Encoder resolution	Encoder
0x6091	1,2	Gear ratio	Encoder
0x6092	1,2	Feed constant	Encoder
0x6098	0	Homing method	Homing tab

Index	Sub	Field	Where
0x6099	1,2	Homing speeds	Homing tab
0x609A	0	Homing acceleration	Homing tab
0x60B2	0	Torque offset	Zero torque here
0x60FF	0	Target velocity	RPDO1

Appendix C — Command-line flags

Flag	Value	Purpose
<code>-a</code> , <code>--auto-connect</code>	—	Connect with default config on startup.
<code>--quit-after</code>	ms	Quit after N ms (headless smoke tests).
<code>--screenshot</code>	path[:ms]	Grab the window to PNG after a delay.
<code>--show-panel</code>	name	Open a dock / dialog (<code>pdo-map</code> , <code>tuning</code> , <code>profile</code> , <code>drive-config</code>).
<code>--show-tab</code>	idx	Select a left-tab (0 Control, 1 Gains, 2 Signal gen).
<code>--only-dialog</code>	—	With <code>--screenshot</code> : capture only the topmost dialog.